

Functional Programming

Lecture 1

Kameliya Golova (she/her)

Constructor University Bremen

Welcome

Plan for today:

- What is functional programming?
- How does this course work?
- Introduction to logic in Lean

What is functional programming?

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The place x is not the same as the value of x.

Imperative languages create places for each variable.

What can this C++ function return?

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    int z = x;  
    y += 1;  
    return x - z;  
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- If x and y are different places, 0.
- If x and y are the same place, 1.

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What is functional programming?

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That's all!

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def x := 0
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- x is not a place, and does not store a value.
- $x := 5$ (i.e. $0 := 5$) does not make sense.

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Idea 1: Functional programming is programming that focuses on *values*.

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Other effects: reading input, throwing an exception, generating a random number, making a web request...

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Idea 2: Functional programming permits, but controls, effects.

Working with Values

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Important distinction:

Value the result of a computation

Expression the program code describing a computation

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The first functional language was the λ -calculus, where these were the only two operations. The λ -calculus is Turing-complete.

In Lean, every expression must have a type, which we denote $t : T$.

For example:

- $5 + 3 : \text{Nat}$
- $\text{fun } n \Rightarrow n + 3 : \text{Nat} \rightarrow \text{Nat}$
- $(\text{fun } n \Rightarrow n + 3) 5 : \text{Nat}$

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Lean is special in that types are also values, and so also have a type.

For example:

- $\text{Nat} : \text{Type}, \text{Nat} \rightarrow \text{Nat} : \text{Type}$
- $\text{Type} : \text{Type } 1$

In the first weeks of the course, we will focus on *propositions*: types whose values are proofs, written $A : \text{Prop}$.

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Idea 3: functions between propositions are implications.

We can see the elements of a proposition $A : \text{Prop}$ as its proofs.

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Idea 4: Types can express propositions.

Idea 5: Terms can express proofs.

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Functional programming language: you can define programs that operate on values.

Interactive theorem prover: you can reason about values and operations on those values.

Together: you can write a program and prove things about it.

Important note: we use Lean **4**.
Lean 3 is very different!

Who uses Lean?

What is functional programming?

Lean is primarily popular amongst mathematicians.

- Large amounts of mathematics are available in [Mathlib](#).
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Lean is also good for cases when you need correctness.

- Amazon used Lean to develop [Cedar](#).
- Microsoft used Lean to verify [post-quantum cryptography in SymCrypt](#).

Logistics



You will learn:

Proofs How to express and prove things in Lean.

Programs How to express computations in Lean.

- And how to prove things about them.

Applications How to go from toy examples to an actual program.

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- A seminar,
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 - More material in a more interactive setting.
- A homework assignment,
 - Organized via the LMS,
 - Submission via GitHub.

Rules and Expectations

Learning a language requires regular, focused work.

In particular:

- The lectures are us constructing things together.
 - Interrupt to ask questions if you have them.
 - Don't expect to be able to follow it while you're on your phone.
- The homeworks are there for you to practice writing it yourself.

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You may use as much AI as you want for the homework assignments. However, you are unlikely to develop the skills needed for the exam unless *you* write Lean code.

Your grade has two components.

50% coursework Autograded weekly homework.

50% exam Written exam, plus up to 10% bonus from quizzes.

The quizzes will be given during the seminars.

You need at least 45% on **each** component to pass (before the bonus).

- Homework repos are distributed via the LMS.
- Check your own score locally: `./grade.sh`.
- Submit by pushing your repo.
- Your grade arrives after the deadline.

First two homeworks:

- HW00 is a trivial exercise to check your setup (due Friday evening).
- HW01 is the first real homework (due Tuesday evening).

Contact

Your options:

- Preferred: message [@kameliya_elektronika_bot](#) on Telegram.
 - I still see your message, but I get a TODO + FAQs get answered.
- Email: kameliya.golova@jetbrains.com
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Teaching assistants:

- Yuliya Karalenka
- Olesia Mutaeva
- Stanislau Palyn



[Announcements](#)



[Chat](#)

If you want to know more about Lean, here are some good resources:

- lean-lang.org
- [Theorem Proving in Lean 4](#)
- [Functional Programming in Lean](#)
- [Learn Lean 4](#)